



Factors that influence the extraction methods of terpenes from natural sources

Ricardo A. González-Hernández^{1,2} · Norma A. Valdez-Cruz^{1,3} · Mauricio A. Trujillo-Roldán^{1,3}

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Abstract

Due to their various applications, terpenes and terpenoids are secondary metabolites of industrial interest. These compounds can be found in a wide variety of organisms from virtually all biological kingdoms. However, it has been identified that various factors in the production process, such as the place and time of harvest, pretreatments, extraction, and purification, generate differences in the profile of terpenes and terpenoids that can be obtained from organisms. The first factor that influences the extraction of compounds is drying since an inappropriate choice of the drying method can lead to the loss of the compounds of interest or the appearance of others that arise due to the decomposition of the original molecules. Similarly, Soxhlet extraction is the most used method in the recovery of terpenoids due to its high extraction efficiency. However, unconventional extraction methods have proven to be effective alternatives for terpenoid recovery, particularly the use of microwave and high-pressure extraction, as well as the use of unconventional solvents, achieving extraction efficiencies equal to or greater than Soxhlet in shorter times. This review will address the differences observed in the extraction of terpenoids and the main factors that cause them, as well as a comparison of the advantages and disadvantages of the various methodologies for drying, extraction, and separation of these compounds.

Keywords Terpenoids · Extraction · Natural products · Separation

Introduction

Terpenes/terpenoids are molecules derived from isopentenyl pyrophosphate (IPP) and are the most abundant secondary metabolites in nature (González-Hernández et al. 2023) (Fig. 1). These compounds are synthesized by a wide variety of plants, fungi, and bacteria, fulfilling various biological functions in the producing organisms. For example, in plants, terpenoids function as a part of the cell membrane, repelling herbivores, controlling pathogens and parasites, attracting pollinators (pheromones), and favoring symbiotic interactions, among others (Langenheim et al. 1994; Pierik et al. 2014). On the other hand, in other organisms, such as bacteria and fungi, terpenoids fulfill functions that include electron transport, cell wall, and membrane formation, chemical defense against predators, and establishment of symbiotic relationships (Calvo et al. 2002; Gershenzon and Dudareva 2007; Cale et al. 2016).

More than 70,000 terpenoid-type molecules have been reported (Helfrich et al. 2019). These compounds are of industrial interest due to their use in the synthesis of fuels, additives, and food flavorings (Mele et al. 2021; Milker

✉ Ricardo A. González-Hernández
raghdz1608@gmail.com

✉ Mauricio A. Trujillo-Roldán
maurotru@iibiomedicas.unam.mx; maurotru@gmail.com;
maurotru@comunidad.unam.mx

Norma A. Valdez-Cruz
adriavaldez1@biomedicas.unam.mx; adriavaldez1@gmail.com

¹ Departamento de Biología Molecular y Biotecnología, Instituto de Investigaciones Biomédicas, Universidad Nacional Autónoma de México, Ciudad Universitaria, 04510 Coyoacán, Ciudad de México, Mexico

² Posgrado en Ciencias Biológicas, Universidad Nacional Autónoma de México, Unidad de Posgrado, Edificio B, 1° Piso, Circuito de Posgrados, Ciudad Universitaria, 04510 Coyoacán, Ciudad de México, Mexico

³ Departamento de Bionanotecnología, Centro de Nanociencias y Nanotecnología, Universidad Nacional Autónoma de México, Km 107 Carretera Tijuana-Ensenada, 22860 Ensenada, Baja California, Mexico

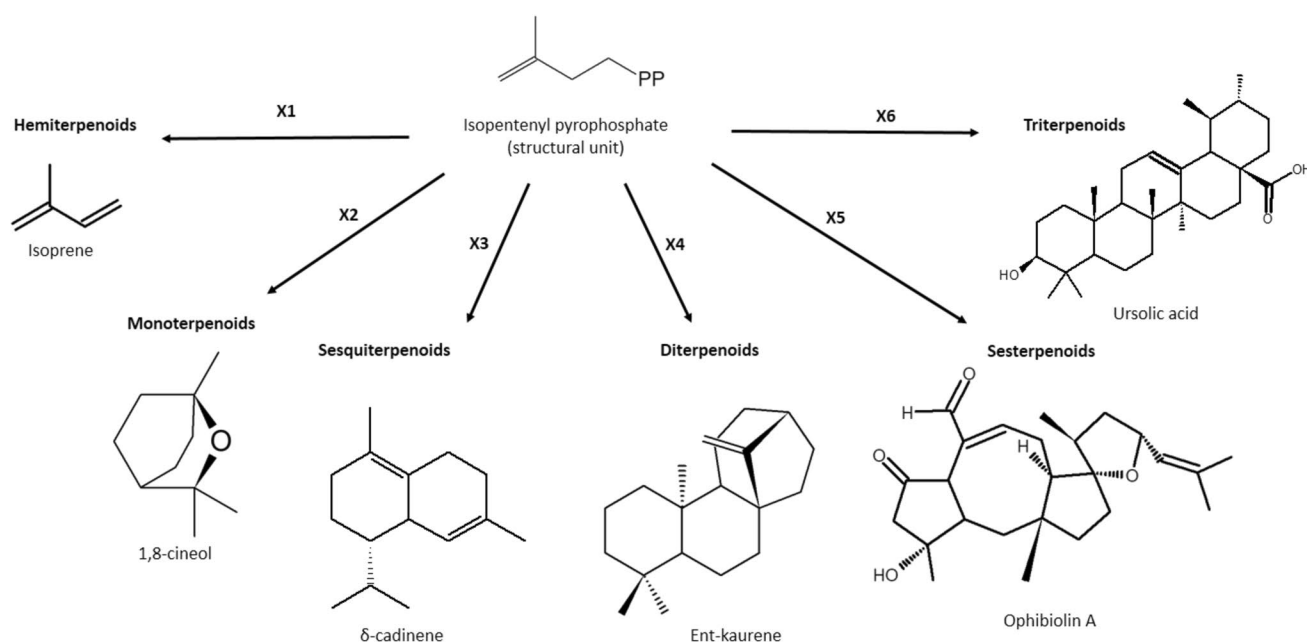


Fig. 1 Examples of terpenoids. These compounds are classified based on the IPP units they contain

and Holtmann 2021), extraction solvents (Dejoye-Tanzi et al. 2012), and their pharmacological properties, including antiviral, antimicrobial, antifungal, cytotoxic and anti-inflammatory activity (Bergman et al. 2019).

A search was carried out for the number of publications made from 2013 to date on terpenoids, extraction, and purification, applying the corresponding Boolean operators for the search criteria: [(terpene OR terpenoid)]; [(terpene OR terpenoid) AND (extraction)] y [(terpene OR terpenoid) AND (purification OR separation)], using search engines Pubmed-ENCBI™, Sciencedirect-Elsevier™ and

Web of Science™ and taking in consideration publications per year (Fig. 2). This search showed an increasing trend in research on the application and obtaining of terpenes/oids, highlighting its importance in various application areas (Fig. 2). For this reason, developing increasingly efficient extraction and purification methods for their application in obtaining these secondary metabolites is of great interest. In this review, an integrative analysis will be carried out on the process of extraction and purifying terpenes/terpenoids and the factors that affect their recovery, giving rise to differences in the recovered compounds (Fig. 3).

Fig. 2 Publications on terpenoids from 2013 until June 2023, based on search engine databases Pubmed-ENCBI™, Sciencedirect-Elsevier™. For the construction of the graph, the topics “terpenes/terpenoids”, “terpenes/oids extraction” and “separation or purification of terpenes/terpenoids” were used

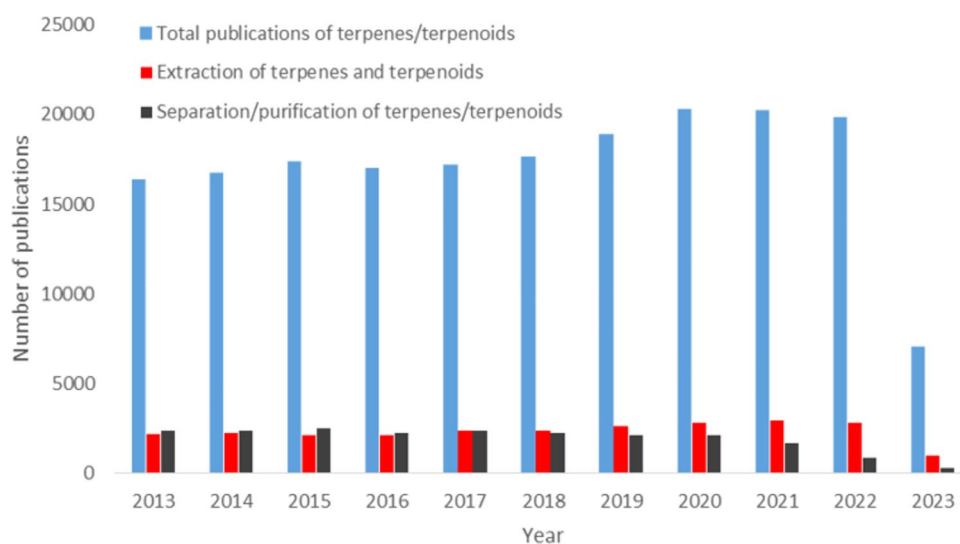
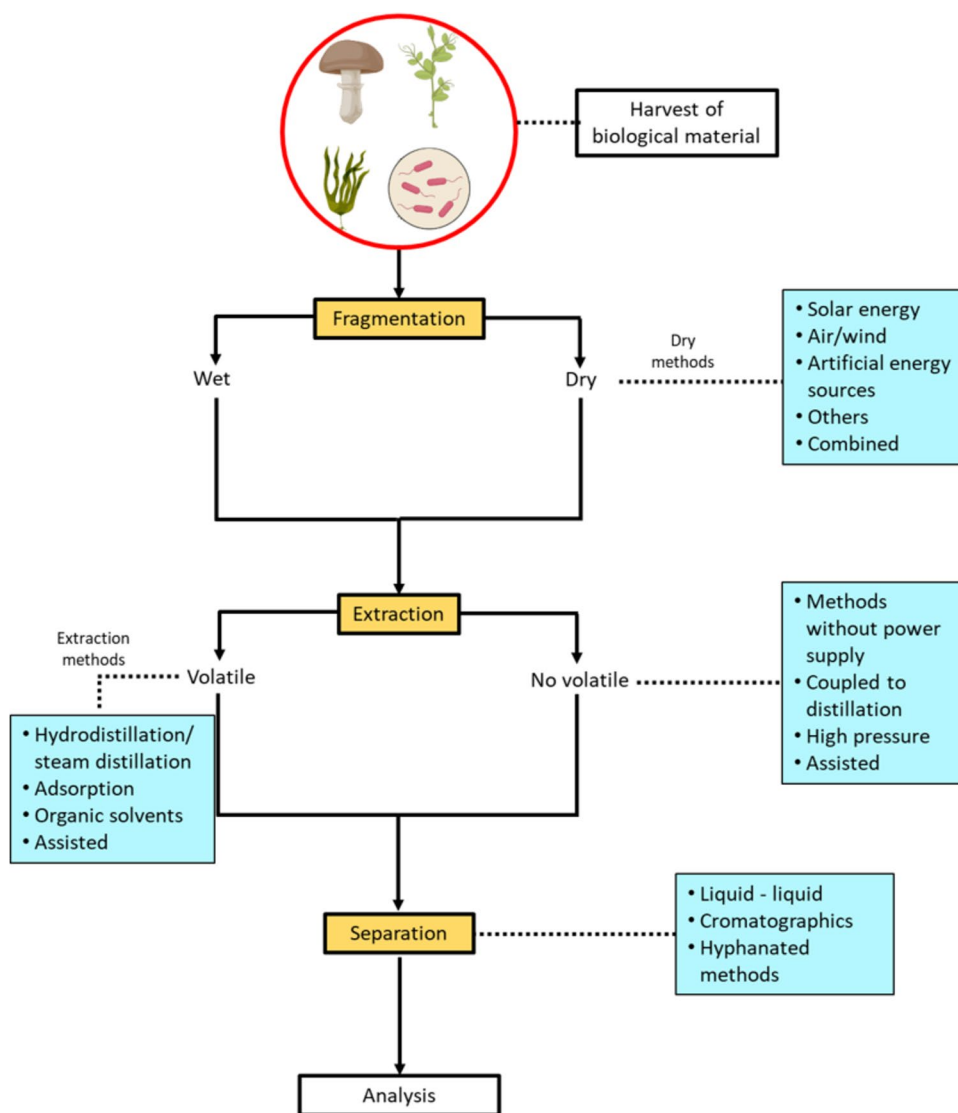


Fig. 3 General scheme for obtaining terpenoids from natural sources. Process steps that can generate variations in the recovered terpenoid profiles are highlighted in orange. In blue, the fundamentals of the different methods of drying, extraction, and separation are mentioned



Preparation of biological material

The first critical step to obtaining enough terpenoids is selecting the most appropriate biological material. Several factors must be considered, including the type of tissue, the age of the producing organism, storage time, maturation time of the compounds, and other treatments before extraction (Mushtaq et al. 2014; Neag et al. 2018; Soib et al. 2020).

Once the biological material has been selected, the first treatment that should be considered is the fragmentation method, since the smaller the particle size, there is greater interaction with the solvent, favoring the extraction of the metabolites, impacting the amount of metabolites recovered (Gião et al. 2009; Hasanov et al. 2019; Rojo-Poveda et al. 2019; Yuliani et al. 2019; Budiastira et al. 2020). Fragmentation (or pulverization) methods are classified into dry and wet methods.

Dry fragmentation methods

In the extraction of low polarity compounds, such as most terpenes and terpenoids, the presence of water in the biological material interferes with the extraction capacity of the solvent used, mainly if it has organic nature (Efthymiopoulos et al. 2019). Therefore, the dry process helps recover low- to intermediate-polarity metabolites. In addition, dry fragmentation is usually less expensive than the wet process, but moisture level control is crucial since conventional grinding methods require low moisture to achieve smaller particle sizes (Lee et al. 2013; Jung et al. 2018; Moon and Yoon 2017). This makes these pulverization methods preferred for extracting active ingredients from natural sources. However, before pulverization, the drying method is decisive in ensuring the quality of the extract obtained. Consequently, for a good recovery of terpenoids from plants or other organisms, the integrity of the plastids or vesicles that store these

compounds must be maintained during the drying process (Ebadi et al. 2015; Singh and Sharma 2015; Thamkaew et al. 2020). Table 1 summarizes the most common drying methods used to extract active principles from natural sources and their advantages and disadvantages.

Effect of drying on the recovery of secondary metabolites

It has been observed that the recovery of secondary metabolites (terpenoids, polyphenols, alkaloids, saponins, glycosides), principally in plants, is affected depending on the drying method used (Omidbaigi et al. 2004; Ferreira and Luthia 2010; Ghasemi Pirbalouti et al. 2013; Roshanak et al. 2016; Saeidi et al. 2016; Raja et al. 2019). For example, the drying process of papaya tree (*Carica papaya*) leaves was evaluated with three different methods: using a convection oven at different temperatures and drying times, cold drying at $-100\text{ }^{\circ}\text{C}$, and drying in the shade. When comparing the properties of the powders obtained from the biological material with the different drying methods, a smaller particle size was reached when drying at $-100\text{ }^{\circ}\text{C}$. However, the best extraction of polyphenolic compounds was achieved when drying was done with a convection oven at $40\text{ }^{\circ}\text{C}$ for 7 h (Raja et al. 2019). A similar phenomenon occurred in obtaining polyphenols from *Camellia sinensis* and guayusa alkaloids using different drying methods, where again the convection oven showed better recovery yields (Roshanak et al. 2016; Manzano-Santana et al. 2018). Manzano-Santana et al. (2018) evaluated the presence of other secondary metabolites: flavonoids, terpenoids, steroids, and mucilages, but quantification was not made to identify whether there were changes in the amount of the extracted molecules depending on the drying method (Manzano-Santana et al. 2018).

Effect of drying on the recovery of terpenes and terpenoids

In the recovery of essential oils, *i.e.* volatile terpenoids, it has been observed that the drying method affects the type and concentration of recovered compounds (Omidbaigi et al. 2004; Ferreira and Luthia 2010; Ghasemi Pirbalouti et al. 2013; Saeidi et al. 2016). In the study carried out by Omidbaigi et al. (2004) using common chamomile (*Chamaemelum nobile*), sun-drying and shade-drying were tested and compared with drying in a convection oven ($40\text{ }^{\circ}\text{C}$). The same components were detected in the essential oil, extracted by hydrodistillation, with the three drying methods, and no differences were found in most of the compounds. However, with drying in the sun, it was possible to obtain a higher concentration of isobutylangelate and 2-methylbutylangelate. In addition, when drying with a convection oven, the recovery of propyl angelate and pinocarvone was minimal (traces), compared to what was

recovered with sun-drying and shade-drying (Omidbaigi et al. 2004). On the other hand, Ferreira and Luthia (2010) evaluated the drying by lyophilization, sun-drying, shade-drying, and with a convection oven for the extraction of artemisinin, artemisinic acid and dehydroartemisinic acid from *Artemisia annua*. For the recovery of artemisinic acid and dehydroartemisinic acid, lyophilization allows significantly higher recovery than the other methods, while for artemisinin recovery, the best method is sun-drying, followed by shade-drying and convection oven (Ferreira and Luthia 2010).

An exhaustive analysis of the essential oil composition of two varieties of basil, purple (*Ocimum sanctum*) and green (*Ocimum basilicum*), was made, comparing fresh extraction against extraction with different drying methods (sun-drying, shade-drying, lyophilization, convection oven, and microwave oven), showing important differences in the concentration and types of terpenoids recovered (Ghasemi Pirbalouti et al. 2013). In the fresh extraction of the purple and green variety, 37 and 26 compounds were detected, respectively, in the essential oil. For the purple variety, when comparing the extracts from the different drying methods, the following results were observed: sun-drying (36 compounds in total; 6 are not detected in the fresh material); shade-drying (38 compounds in total; 10 not detected in fresh material); convection oven at $40\text{ }^{\circ}\text{C}$ (29 compounds in total; 3 not detected in fresh material); convection oven at $60\text{ }^{\circ}\text{C}$ (13 compounds in total; no differential compound was detected); microwave oven at 500 W (36 compounds in total; 6 not detected in fresh material); lyophilization (27 compounds in total; 4 not detected in fresh material). When comparing the results of the different drying methods, it is observed that there is one compound that is generated exclusively by sun-drying and with microwave oven at 500 W (α -terpinene), 2 exclusive compounds by shade-drying (veridiflorol and menthol), and one exclusive compound when lyophilized (Δ -3-carene). Likewise, the least effective drying method in this study was the use of the convection oven at $60\text{ }^{\circ}\text{C}$, which denotes the need to also evaluate different parameters when using the different drying methods. Regarding the green variety, a similar phenomenon is observed, with differences in the recovery of compounds based on the drying method used (Ghasemi Pirbalouti et al. 2013).

It is evident that the selection of the drying method is a crucial step to have an adequate recovery of active principles and particularly terpenoids. The selection of the drying method must be based on the chemical characteristics of the compounds of interest, such as volatility, reactivity, and susceptibility to thermal or oxidative decomposition. However, when the compound profile of the biological material is unknown, it is advisable to evaluate different drying methods to choose the most appropriate for the purposes of the study.

Table 1 Comparison of drying methods in natural products

Drying method	Description	Advantages	Disadvantages	References
Sun-drying	It consists of placing the biological material outdoors under the sun's rays. It should be placed in thin layers, to facilitate drying and prevent the development of pathogens	A large amount of biological material can be processed at the same time, with a low energy cost	Requires large tracts of land Susceptibility to contamination or herbivory Unsuitable for photosensitive molecules Depends on weather conditions (rain)	Janjai and Bala (2012), Orphanides et al. (2013, 2016)
Solar-assisted drying	It is a closed system made up of two chambers, which achieves desiccation by convection Uses solar energy to heat air and channel it toward the sample (in a chamber) using fans or pumps It has heat exchangers that allow the condensation of water in the second chamber	Reduces drying time by up to 50% compared to drying in the sun Maintains the quality of the essential oil Contaminant sources are controlled Requires less space to install the system, compared to sun-drying and shade-drying	Depends on weather conditions (availability of sunlight) Requires trained personnel for maintenance	Janjai and Tung (2005), El-Sebaï and Shalaby (2012), Tiwari (2016), Rabha et al. (2017), Amer et al. (2018), Thamkaew et al. (2020)
Shade-drying	The biological material is placed indoors with access to air flows, which carry moisture Take advantage of the heating of the air by the sun, to capture more humidity	Contaminant sources are controlled Metabolites are better preserved than with other methods It is applicable for obtaining photosensitive molecules	Drying times are very long Depends on ambient humidity, material can be easily rehydrated	Capecka et al. (2005), Sharma et al. (2009), Orphanides et al. (2013, 2016), Thamkaew et al. (2020)
Hot air drying	The method is based on the principle of solar-assisted drying. However, it uses a heating system for the air, not dependent on sunlight	Contaminant sources are controlled Requires less space to install the system, compared to sun-drying and shade-drying Temperature and air velocity can be controlled (optimisable) Useful for recovering thermolabile compounds Does not depend on environmental conditions	The system involves considerable energy expenditure Requires trained personnel for maintenance If the parameters are not properly adjusted, there may be losses of volatile compounds An oxidizing environment can be generated that alters the metabolites of interest	Fennell t al. (2004), Doymaz (2006), Antal (2010), Orphanides et al. (2016), Thamkaew et al. (2020)
System assisted with heating pump	The system is based on air dryers. In this case, a heating pump is coupled to the desiccator By incorporating a cooling system for humid air, the system can be made more efficient by using completely dry air	Air temperature can be regulated By using completely dry air, drying time is decreased (~48%) Compared to other convective drying methods, this method maintains higher material quality Low energy consumption	There is a lack of studies to evaluate the effect on the conservation of volatile compounds High installation and maintenance costs	Best et al. (1996), Islam and Mujumdar (2004), Fatouh et al. (2006), Patel et al. (2012), Thamkaew et al. (2020)

Table 1 (continued)

Drying method	Description	Advantages	Disadvantages	References
Fluid bed drying	The process is like drying with hot air. The main difference is that the biological material is layered, and the hot air is flowed at high velocity	The method provides short drying times Using heat exchangers gives you all the combined advantages of methods using hot air drying and heat pump Easy handling It can be adapted with other methods such as microwave and infrared	The pressure drops due to the maintenance of the flow velocity, entails a high energy consumption The homogeneity of the final product is not ensured Some volatile compounds can be lost with this technique (the process must be optimized) Heating can affect the quality of the final product	Singh and Kingsly (2008), Buchailot et al. (2009), Calin-Sánchez et al. (2020), Thamkaew et al. (2020)
Infrared radiation drying	The biological material is placed in a chamber where infrared radiation ($\lambda = 0.8\text{--}1000\text{ }\mu\text{m}$) is incident on it	Simplicity and speed of the process Can be coupled with other drying methods Maintains the quality of the metabolites to extract It is a method that makes heating more efficient at a low energy cost	If heating uniformity is not maintained, it can damage the metabolites of interest (thin layers) Large amounts of biological material cannot be processed at one time Potential fire risk	Boudhrioua et al. (2009), Ashtiani et al. (2017), Calin-Sánchez et al. (2020), Thamkaew et al. (2020)
Microwave drying	A system is used that contains an oven in which electromagnetic radiation is incident with a frequency between 300 MHz and 300 GHz. For this, klystron or magnetron oscillators are required, which convert electrical energy into microwaves	A larger volume of sample can be processed at one time, compared to infrared Short drying times Easy handling It is one of the methods that maintain a low energy cost	High initial cost Not suitable for volatile compounds Alters the texture of biological material	Soysal (2004), Orsat et al. (2006), Heindl and Müller (2007), Orphanides et al. (2016), Chua et al. (2019), Calin-Sánchez et al. (2020), Thamkaew et al. (2020)
Ultrasonic drying	High frequency vibrations are induced in the material (20 kHz – 10 MHz)	Short drying times Maintains the integrity of most compounds Does not require heat treatment Can be coupled with other drying methods. Low energy cost	Damage to biological material occurs Acoustic waves can be quickly attenuated	Jambrak et al. (2007), Teng et al. (2019), Calin-Sánchez et al. (2020), Thamkaew et al. (2020)
Vacuum-assisted	It uses a system consisting of compartments, connected to a vacuum pump	Achieves obtaining dry material in short times (12–48 h) It is useful for obtaining photosensitive and thermosensitive molecules Can be coupled with other drying methods The water recovered from the biological material can be used for other purposes Low energy cost	Poor process standardization can degrade the quality of essential oils	Parikh (2015), Calin-Sánchez et al. (2020)

Table 1 (continued)

Drying method	Description	Advantages	Disadvantages	References
Lyophilization	It consists of freezing the biological material and placing it in a chamber to reduce the pressure. This allows the sublimation process to eliminate the water present	Instantly stops enzymatic and oxidation reactions of molecules Maintains the porous structure in the material	The process is very expensive due to the requirement of specific facilities The process is slow There are significant losses of the most volatile compounds (essential oil)	Díaz-Maroto et al. (2002), Antal (2010), Pirbalouti et al. (2013), Rahimmalek and Goli (2013), Szychowski et al. (2018), Calín-Sánchez et al. (2020), Thamkaew et al. (2020)
Instant controlled pressure drop (DIC)	The biological material is placed in a chamber where differential pressure "pulses" can be performed It is maintained at high pressures (4–5 atm) for short periods of time (~20 s). Subsequently, the pressure is suddenly dropped (0.4–0.5 atm), and the process is repeated cyclically	The process causes expansion of materials texture without destruction of the biological material The process is fast and maintains the quality of the extracts	The process and maintenance are expensive	Mounir et al. (2011), Allaf et al. (2013)
Supercritical fluid drying	The material is placed in a chamber where supercritical fluid (usually CO ₂) is passed through it at low pressure for several days (at least 4)	Useful for recovering thermolabile compounds Inhibits the proliferation of aerobic microorganisms Inhibits oxidation reactions of metabolites It is compatible with ultrasonication	Requires special facilities to maintain supercritical fluid Studies are lacking to evaluate the impact on the quality of the extracts when using it	Busic et al. (2014), Michelino et al. (2018), Thamkaew et al. (2020)

Wet fragmentation

The use of wet grinding is useful when it is intended to recover high-polarity molecules and water or solvents miscible with it will be used, such as methanol, ethanol, and acetone, among others. The terpenes that form saponins (glycosylated) have a relatively elevated polarity, which is why the choice of this grinding method is useful for recovering them. In addition, it has been observed that for some plant products, moistening the material favors grinding, and currently, ultrafine grinding systems allow the wet process to reach particle sizes smaller than 1 μm (Pan and Tangratanaalee 2003; Zhao et al. 2009; Karinkanta et al. 2018). One of the advantages of wet grinding methods is that the quality of the final product is preserved, which implies less loss of compounds (Tong et al. 2015; Jung et al. 2018). Additionally, some wet milling methods can be directly coupled to extraction, adding the solvent as in the case of ultrasonication (Budiastra et al. 2020), which will be discussed later. Due to the high-frequency vibrations generated by ultrasonication, pores can be generated in the cells of the biological material or even lysed. The use of liquid nitrogen to freeze the samples and be able to pulverize them without the need to reduce the humidity is still valid due to its versatility since it allows for stopping the enzymatic reactions that are still active within the biological material (Mushtaq et al. 2014; Jiang et al. 2016; Neag et al. 2018).

In addition to these methods, there are other wet grinding techniques that use mills or disintegrators specifically designed for the process. With wet mills, particle sizes in the order of nanometers have been achieved using polymeric media, but the energy cost is still very high (Bilgili et al. 2004; Zhao et al. 2009; Gao et al. 2020). In these mills, the pulverization principle is due to a repeated process of compression and impact of the particles of biological material, classifying them into five main groups: roller mills, ball mills, air jet mills, impact mills, and those of another type (which includes the rest of the mills Yokoyama and Inoue 2007; Karinkanta et al. 2018; Gao et al. 2020).

Wet fragmentation for the recovery of terpenes and terpenoids

The use of wet milling has allowed a significant increase in the recovery of terpenoids of intermediate polarity (hydroxylated and carboxylated), glycosylated (saponins), and terpenic alkaloids, in addition to other more polar compounds such as polyphenols and alkaloids, since by reaching such small particle sizes, the interaction with polar extraction solvents is better (Wu et al. 2021). Although wet grinding is less used than dry grinding to obtain natural products, wet methods have good application potential, particularly in recovering high-polarity metabolites.

As for the fragmentation methods, it can be concluded that it is necessary to carry out specific tests to optimize the processing of plant material based on the type and polarity of compounds to be recovered. In addition, the cost–benefit must be considered when selecting a dry grinding method (considering the most convenient type of drying), or wet grinding methods, since the latter are usually more expensive in economic and energy terms.

Extraction methods for the recovery of terpenes and terpenoids

Extraction is a key step in the recovery of secondary metabolites since the process can be directed to the recovery of most compounds (exhaustive extraction) present in the biological material or only to a particular group (non-exhaustive extraction). For this, an appropriate solvent selection is necessary because, in addition to thinking about the type of compounds to be recovered, compatibility with the extraction method must be ensured (Pawliszyn 2012). Since terpenes contain only carbon and hydrogen in their structure, these molecules can be recovered using fewer polar solvents. Meanwhile, terpenoids, which have polar functional groups (hydroxyl, carboxyl, and even amino) or can even be glycosylated (saponins), intermediate- to high-polarity solvents are commonly used; for example, chloroform and methanol (Tongnuanchan and Benjakul 2014; Jiang et al. 2016; Stratakis and Koidis 2016; Aziz et al. 2018). Before starting the extraction, the first question is whether the extraction will be directed when the structure of the compound of interest is known; or undirected when there is no information about the main compounds of the biological material. This question is essential to select the most appropriate method and solvents (or a mixture thereof) for the extraction. In addition to this, in the case of terpenoids, it is necessary to consider the factor of the volatility of the compounds (Tongnuanchan and Benjakul 2014; Jiang et al. 2016; Aziz et al. 2018). Conventional methods for obtaining volatile terpenes/terpenoids include steam stripping, maceration, and adsorption methods (Silva et al. 2020; Pawliszyn 2012; Azmir et al. 2013; Tongnuanchan and Benjakul 2014; Jiang et al. 2016; Stratakis and Koidis 2016; Aziz et al. 2018). For non-volatile terpenoids, maceration, decoction, infusion, reflux, or Soxhlet methods are commonly used (De Vos et al. 2007; Mushtaq et al. 2014; Jiang et al. 2016).

When the profile of compounds presented by the biological material is unknown, an undirected analysis is used, which focuses on recovering most of the secondary metabolites present in the sample and characterizing them. In these cases, consecutive extractions are used, using different solvents of increasing polarity and mixtures of these (Mushtaq et al. 2014; Wu et al. 2014).

Extraction of volatile terpenes and terpenoids

Volatile terpenes/terpenoids are generally five-carbon structures (hemiterpenes/terpenoids), ten carbons (monoterpenes/terpenoids), and 15 carbons (sesquiterpenes/terpenoids) molecules (Abbas et al. 2017). For the recovery of volatile terpenoids, the methods are based on carrying the volatile molecules with the vapors of a solvent or using materials that act as traps based on the principle of adsorption.

Steam distillation

Steam distillation techniques are commonly used for volatile compound extraction, which is based on Dalton's law of partial pressures and immiscibility with water. To date, four variants of this method are known: hydrodistillation, direct steam stripping, indirect steam stripping, and a technique called distillation with water-bubble distillation (Table 2). One of the advantages of these methods is the use of water, considerably reducing the generation of harmful waste for the environment, and due to their design, they are easily scalable and do not require complex technologies (Aramrueang et al. 2019; Azmir et al. 2013; Fitri et al. 2017).

In the particular case of distillation by water-bubbling, it is a relatively new method whose first report dates from 2017; it has been observed that this is a non-exhaustive extraction method. It has been evaluated in the extraction of patchouli (*Pogostemon cablin*) essence, showing a higher proportion of patchoulol, compared to stripping with steam and water, which gives the extract a higher quality (Fitri et al. 2017).

Adsorption-based methods

These adsorption techniques were developed in the 40 s, and due to their practicality, new and more efficient methods with a greater capacity to capture volatile compounds continue to be developed (Yang et al. 2013; Zhang et al. 2023). They are based on the principle of adsorption of volatile compounds to a stationary phase that functions as a trap. These methods are classified as static: solid phase extraction (SPE), solid phase microextraction (SPME); and dynamic: purge and capture, adsorbent trap, and its variations (Pawliszyn 2012). Static headspace methods require a closed system that allows the transfer of volatile compounds from the sample to the adsorbent and requires reaching the equilibrium of saturation of the adsorbent with volatile compounds. Instead, dynamic headspace methods rely on using inert gases to accelerate the extraction of volatile compounds into the adsorbent. In most cases, with these methods, the saturation of the adsorbent is not reached, but it allows faster recovery of a greater variety of compounds (Kremser et al. 2016).

Among the main advantages of adsorption, methods are mainly at an analytical scale to obtain compound profiles, increasing extraction efficiency by reducing organic solvents (Pawliszyn 2012; Kremser et al. 2016). However, one of the main drawbacks is that these methods are, up to now, not useful for industrial recovery of compounds since they depend on the saturation of the adsorbent.

Recovery of terpenes and terpenoids with steam distillation and adsorption methods

In the recovery of volatile terpenoids, differences have been observed in the type of compounds that are obtained when using steam distillation methods and those of adsorption. In the study carried out by Niculau et al. (2020) on the extraction of volatile terpenoids from geranium, it was shown that there are differences in the profile of compounds recovered with hydrodistillation and two dynamic adsorption methods for the headspace. With hydrodistillation, no monoterpenes were recovered, only monoterpenoids, sesquiterpenes, and esters, of which the main ones are linalool and citronellyl formate, in a higher proportion compared to adsorption methods. While with the dynamic headspace adsorption methods, it was possible to recover compounds from the group of monoterpenes and monoterpenoids, sesquiterpenes and esters, mainly citronellol, geraniol, 6,9-guaidiene, geranylthiglate, and citronellyl formate. Here it is notorious that adsorption methods allow the recovery of a greater variety of compounds but in smaller quantities.

When selecting the adsorption methods, it is essential to define the type of phytochemical characterization that is desired to be achieved. Non-exhaustive methods such as SPME and headspace dynamics are good choices for identifying major components, while the solid phase extraction and purge trap methods take longer but allow the complete profile of compounds to be obtained (Kremser et al. 2016; Diez-Simon et al. 2020).

Extraction with organic solvents

Volatile compounds can also be recovered with methods using organic solvents. However, these methods are not specific and recover volatile and non-volatile molecules. These methods will be discussed in the non-volatile terpenoid extraction section.

Extraction of non-volatile terpenoids

Non-volatile terpenoids are made up of di-, ses-, tri-, tetra-, and polyterpenes/terpenoids and glycosylated mono- and sesquiterpenoids. To obtain these compounds, solvents of different polarities are used; however, there are different strategies for obtaining nonpolar compounds (Table 2).

Table 2 Comparison of conventional terpeneoid extraction methods

Terpenes types	Extraction type	Variant	Description	Advantages	Disadvantages	References
Volatiels	Hydrodistillation- Steam distillation	Hydrodistillation	The biological material is immersed in water and brought to a boil. The water vapor carries the compounds	The equipment is the easiest to assemble and manipulate Greater penetration than steam drag for very compact samples (powders) It can drag molecules that are not so volatile, but of low molecular weight	Thermolabile compounds are susceptible to decomposition Much slower compared to other methods It can carry unwanted compounds other than terpenoids	Silva et al. (2020), Azmir et al. (2013), Fitri et al. (2017), Aramrueang et al. (2019)
		Direct steam distillation	Water vapor is passed through the plant material to drag the compounds in the same container, without submerging the biological material	It is a less aggressive method than hydrodistillation It is selective for very volatile molecules The process is easy to scale Solvent can be reused in several cycles	The assembly of the system is relatively expensive Recovery is not efficient in very compact materials or with high hardness	
		Indirect steam distillation	Plant material is in a container separate from the steam source. Subsequently, it is saturated with water vapor, coming from another container, to drag the compounds	Increases the penetration of vapors to extract volatile compounds (higher performance) More reproducible extraction profiles The process is easy to scale Solvent can be reused in several cycles	It is slower than direct steam drag There are steam pressure losses in the process (slower) The assembly of the system is relatively expensive The energy cost is higher	
		Water-bubble distillation	It is a relatively new technique Combines the use of a kettle still with a stirrer to generate bubbles that penetrate biological material	It mainly recovers the molecules that have higher volatility, compared to extraction by indirect steam stripping. Can be easily scaled Large sample volumes can be processed at one time (depends on containers)	It leads to higher energy expenditure The process is slower than with steam stripping Not useful for recovering all volatile compounds (non-exhaustive)	

Table 2 (continued)

Terpenes types	Extraction type	Variant	Description	Advantages	Disadvantages	References
Adsorption		Purge and trap	This is a dynamic method of recovery of volatile compounds An inert gas drags the volatile compounds from the extract matrix to be subsequently trapped in an adsorbent	The recovery of the main components is achieved, in enriched fractions Different types of adsorbents can be used which can make the recovery of compounds more selective	As it is not an exhaustive method, not all the volatile components of the sample are recovered Its application is limited to analytical studies, since its use on a large scale is not profitable The use of inert gases can make the process more expensive	Richter and Schellenberg (2007), Kremser et al. (2016), Jelen et al. (2021)
		Solid phase extraction	It is a static method of recovery of volatile compounds. It is required to reach the saturation equilibrium of the resin with the volatile compounds of the headspace	It is possible to obtain a more complete profile of the volatile terpenoids Selective resins can be chosen for specific groups of compounds It has a higher retention capacity of compounds compared to other methods	The method is slower compared to purge-trap and solid phase microextraction	
		Solid phase microextraction	It is a non-exhaustive automated method. The extract is flowed through the resin and a pre-equilibration is established, like that of solid phase extraction	Does not use solvents Short analysis time Automated Low cost, compared to other methods	The saturation process is usually the limitation in this method Interference by molecules suspended in the extract is common	
Non-volatile	Organic solvents	Without heating	Maceration	It does not require energy consumption. Multiple samples can be processed at once Large quantities of myestra can be processed Process temperature can be adjusted Useful for thermolabile compounds It can be coupled to continuous agitation systems	The extraction process is very long Consumes large volumes of solvent Depending on the solvent, harmful residues can be generated for the environment	Jiang et al. (2016), Zhang et al. (2018), Ternelli et al. (2020), Truong et al. (2021)

Table 2 (continued)

Terpenes types	Extraction type	Variant	Description	Advantages	Disadvantages	References
	Heating	Percolation	The biological material is placed on a filter and solvent is flowed through it. The process can be carried out at different temperatures and pressures, to make it more selective	Does not require higher energy expenditure Large amounts of biological material can be processed at one time Extraction temperature can be adjusted With the help of vacuum systems, the process can be accelerated	Large amounts of solvent required	Zhang et al. (2018), Chemat et al. (2020), Ternelli et al. (2020), Truong et al. (2021)
		Decoction	It consists of placing the biological material immersed in the solvent and bringing it to a boiling point. Non-flammable solvents should be used, mainly water	It is useful for processing difficult-to-pulverize samples Increases solvent penetration Waste is usually low risk	Because water is commonly used, nonpolar terpenoids are not recovered Not suitable for thermolabile substances Requires a lot of energy	Zhang (2023), Liu et al. (2018), Zhang et al. (2018), Fadel et al. (2020), Jedidi et al. (2021)
	Distillation coupled	Reflux	The biological material is placed immersed in the solvent and heated to the boiling point, then the solvent returns to the extraction matrix and the process ends when osmotic equilibrium (solvent saturation) is reached	Requires less amount of solvent than mashing and percolation Combines the advantage of decoction, using organic solvents	Solvent saturation is reached prior to recovery of all compounds The method requires an appropriate selection of solvent to recover thermosensitive molecules Requires relatively high energy expenditure	Zhang et al. (2018)
		Soxhlet	Continuous and exhaustive extraction method (cyclical). Uses the reflux and percolation principle in each cycle	Low volume of solvent required compared to maceration or percolation Allows complete recovery of soluble compounds in the solvent (does not saturate)	For the recovery of molecules sensitive to thermal decomposition, a precise choice of solvent is required For efficient extraction, long periods of time (multiple cycles) are required Requires relatively high energy expenditure	Danlami et al. (2014), Azwanida (2015), Jiang et al. (2016), Zhang et al. (2018), Tzanova et al. (2020)

Methods without heating required

The main methods that do not require heating are cold maceration and percolation; regarding the process, they are the simplest and have the lowest energy consumption compared to the others. Maceration is one of the oldest extraction methods, but due to its practicality, because it does not require energy expenditure or greater manipulation, it is still widely used today (De Vos et al. 2007; Jiang et al. 2016; Zhang et al. 2018; Ternelli et al. 2020; Truong et al. 2021). Percolation is another method that can be carried out without heating, which is still widely used; it is based on making the solvent flow through the biological material by gravity action. In addition, the solvent can be used at different temperatures and can even be recirculated until saturated and later use as fresh solvent depending on the requirements of the process (Zhang et al. 2018; Chemat et al. 2020; Ternelli et al. 2020; Truong et al. 2021). In both cases, the selection of the extraction solvent is decisive for obtaining the extracts since particular groups of compounds or more general extractions can be obtained from this. For the recovery of terpenoids without polar functional groups, the use of low–intermediate-polarity solvents such as hexane, dichloromethane, and chloroform, among others, is suggested, while for terpenoids with polar functional groups, the chloroform–methanol mixture (1:1) is the most widely used option (De Vos et al. 2007; Mushtaq et al. 2014; Friesen et al. 2015). For terpenoids bound to alkaloids or carbohydrates, extraction systems containing acidified water mixed with some alcohol, such as methanol or ethanol, are commonly suggested (De Vos et al. 2007; Mushtaq et al. 2014).

In addition to conventional solvents, maceration and percolation are compatible with ionic liquids at room temperature (Bogdanov 2017; Friesen et al. 2015). This is extremely important for obtaining secondary metabolites by these techniques since ionic liquids have shown to have a high extraction capacity, and by not requiring the supply of external energy, the process can become very profitable.

Methods that use heating

One of the limiting factors in the extraction process is the permeability of the biological material to solvents. If the solvent is inefficient in penetrating the cell wall of the organism, the recovery of secondary metabolites will be deficient. For this reason, heating liquid solvents to decrease their viscosity favors the ability to penetrate plant material by increasing fluidity, consequently achieving greater extraction efficiency.

The decoction method consists of placing the plant material immersed in water (generally) and subjecting it to an increase in temperature until it reaches the boiling point. Using water as a solvent and being a very aggressive heating

method, it is not recommended to extract terpenoids or saponins (Zhang et al. 2018). However, some reports mention the recovery of some terpenoids, terpenic alkaloids, and saponins by this technique, but these compounds must be resistant to thermal decomposition (Liu et al. 2018; Zhang et al. 2018; Fadel et al. 2020; Jedidi et al. 2021; Wang et al. 2021). In addition to decoction, infusion can be used, a milder hot extraction method, where the biological material is placed in the hot solvent (equivalent to a high-temperature maceration).

On the other hand, extraction methods with solvent distillation are more efficient than decoction and methods without heating (Zhang et al. 2023; Zhang et al. 2018; Tzanova et al. 2020; Wei et al. 2020). Direct reflux extraction consists of a container coupled to a condensation system that allows the solvent to return to the extraction matrix after reaching boiling (Jiang et al. 2016; Zhang et al. 2018). The Soxhlet method is one of the most efficient for the extraction of secondary metabolites because it combines the principles of reflux and percolation extraction, in addition to keeping the biological material separate from the extract, so that the fresh solvent, when distilled, does not reach saturation point (Zhang et al. 2018; Tai et al. 2029; Tzanova et al. 2020). This makes the Soxhlet method one of the most exhaustive processes for the recovery of compounds of natural origin, and for this reason, it is one of the most used methods in the recovery of secondary metabolites; however, it maintains some disadvantages, such as the processing time and the volume of solvents required (Danlami et al. 2014; Azwanida 2015; Jiang et al. 2016; Zhang et al. 2018).

Effect of the extraction method on the recovery of non-volatile terpenoids

In the recovery of terpenoids, the different conventional methods have shown different recovery capacities, and this process also depends on the appropriate selection of the solvent to improve the yields (Jiang et al. 2016; Jamal et al. 2018; Zhang et al. 2018; Al Rashid et al. 2019; Kaur et al. 2019; Singh and Suryanarayana 2019; Chemat et al. 2020). In the study carried out by Jamal et al. (2018), the capacity of various extraction methods for the recovery of ursolic acid was compared from *Lantana camara* leaves (shrub verbena); it was observed that the Soxhlet method was more efficient than maceration, direct reflux and even extraction assisted by ultrasonication, as well as different solvents (methanol, ethanol, acetone, and chloroform), of which methanol was the one with the highest extraction yield (Jamal et al. 2018). Likewise, it was observed that for the extraction of total terpenoids, equivalent to lupeol, from *Diospyros melanoxylon* leaves, Soxhlet presents a better recovery than maceration using the water–methanol mixture (95:05) for the extraction (Al Rashid et al. 2019). For the recovery of terpenoids

(ursolic acid, oleanolic acid, lupeol, and betulinic acid) from *Swerita* sp. species, with the Soxhlet method, a higher terpenoid recovery was achieved using aqueous ethanol (50%) or methanol as solvent, compared to reflux extraction. While using ethyl acetate as a solvent, better recovery of ursolic, betulinic, and lupeol acids was achieved with reflux extraction (Kaur et al. 2019). These studies show that Soxhlet is the most efficient of the conventional methods, and for this reason, it is still widely used to obtain terpenes/terpenoids and other secondary metabolites.

Unconventional extraction methods

Currently, in addition to conventional methods for obtaining volatile and non-volatile terpenes/terpenoids, more efficient extraction processes are used, including ultrasonication-assisted (UAE), microwave-assisted (MAE), vacuum-assisted (VAE), supercritical fluid (SFE) extraction methods, and pressure solvent extraction (PLE), ultrahigh pressure (UPE), solid phase microextraction, among others that will be addressed in this review (Tongnuanchan and Benjakul 2014; Jiang et al. 2016; Stratakos and Koidis 2016; Xi 2017; Otero et al. 2018; Neag et al. 2018; Mahato et al. 2019).

Ultrasonic-assisted extraction (UAE)

This technique is based on using ultrasonic acoustic vibrations to break the cells of the plant material and to be able to release its content into the solvent. For this, a generator, a transducer, a signal amplifier, and a probe that will contact the sample are required (Osorio-Toblón 2020). The process can be coupled to all conventional extraction methods, improving efficiency and reducing the use of solvents and extraction time, with low handling cost (Tongnuanchan and Benjakul 2014; Safdar et al. 2016; Deng et al. 2017; Xi 2017; Osorio-Toblón 2020). Due to its advantages, ultrasonic-assisted extraction processes are increasingly used to extract secondary metabolites (Safdar et al. 2016; Deng et al. 2017; Xi 2017; Osorio-Toblón 2020). To obtain terpenes and terpenoids with UAE, the choice of the most appropriate solvent is crucial for the process since this method is coupled to conventional extraction equipment (Cantarino et al. 2020). Regarding the recovery of molecules in essential oils, a study carried out by Pudziuvėlyte et al. (2018), using *Elsholtzia ciliata* (crested late summer mint) as a model, observed that the use of UAE (10–60 min) has a better recovery of volatile terpenoids than maceration with constant agitation (24 h). Also, this presented recovery of non-volatile terpenoids such as phytol and squalene; the rest of the profile of compounds is very similar to that obtained with percolation for 48 h. Likewise, Palmieri et al. (2020) compared the profile of essential oil compounds from thyme (*Thymus vulgaris*), hemp (*Cannabis sativa*), and coriander

(*Coriandrum sativum*), using different extraction methods and absolute ethanol as solvent. It was observed that the UAE method (15 min) has a higher recovery of compounds than maceration (30 days) in the three plants. In contrast, compared to the Soxhlet method (extraction for 6 h) in thyme essential oil, UAE achieved a higher recovery in proportion of carvone, α -terpineol, linalool oxides, β -pinene oxide, (Z)-p-menth-2-en-1-ol, sabinene o-cymene hydrate. Still, the Soxhlet extracted a higher proportion of carvacrol. In the case of hemp, Soxhlet proved to be more efficient for extraction than UAE, since a greater profile of compounds was obtained, having a higher proportion of D-limonene, 1,8-cineole, terpinolene, linalool, caryophyllene, α -bergamotene, selin-3,7,(11)-diene, as well as compounds not recovered by UAE including α -pinene, α -thuyene, β -myrcene, guaia-3,9-diene; for its part, the UAE had a greater accumulation of β -ocimene, β -pinene, cis-p-ment-2,8-dien-1-ol, aromandendrene, α -selinene, cis α -bisabolol. Finally, from UAE coriander, he was able to recover 1,8-cineole, which was not detected in the Soxhlet extraction, as well as a greater accumulation of 2-carene, β -myrcene, γ -terpinene, α -terpineol (Palmieri et al. 2020).

As non-volatile terpenoids, the recovery of ursolic acid from *Lantana camara* was evaluated, the UAE using acetone and chloroform as solvents has a higher recovery efficiency than maceration and is similar to that obtained with direct reflux but does not achieve the efficiency that Soxhlet (Jamal et al. 2018). In the comparison made by Al-Rashid et al. (2019), UAE showed a better extraction capacity of total terpenoids (equivalent to lupeol), than maceration with all the solvents evaluated and similar to the extraction obtained by Soxhlet (for 25 h), but in less time.

In a study focused on the extraction of terpenoids from *Andrographis paniculata*, it was observed that the ultrasonication-assisted maceration method has greater efficiency for the recovery of andrographolide and 14-deoxy-11,12-dihydroandrographolide, compared to maceration with elevated temperature (80 °C) (Chia et al. 2020). As can be seen, ultrasonic-assisted maceration manages to achieve, in several studies, similar results to extraction by the Soxhlet method; however, the most significant advantage that UAE has is the reduction in extraction times (Djenni et al. 2013).

Microwave-assisted extraction (MAE)

This is a very efficient technique for the recovery of secondary metabolites due to its ability to heat the solvent molecules without the formation of temperature gradients, which facilitates the diffusivity of the solvent and rupture of the cellular structures of the biological material (Veggi et al. 2012; De Calle and Costas-Rodríguez 2017; Yahya et al. 2018; Osorio-Tobón 2020). The technique supplies low-frequency electromagnetic radiation (2.45 GHz) to the

biological sample in contact with the solvent. Such as UAE, microwave-assisted extraction can be coupled to conventional extraction methods with the advantage that extracts are obtained in less time (Jones et al. 2001; Mushtaq et al. 2014; Cantarino et al. 2020; Osorio-Tobón 2020). There are two modalities of application of the MAE: the first is when the radiation is delivered in a focused manner (single mode) and is used at ambient pressure in open systems, and the second is known as multimodal and occurs when radiation is supplied in a non-focused manner, which occurs at high pressures and in closed systems (Osorio-Tobón 2020). The choice of microwave operation modality will depend on the quantity, density, and thickness of the sample. When samples are in thin layers, the most appropriate modality is single mode, while in large and thicker samples, the multimode is the best choice.

The coupling of the MAE method with conventional techniques for obtaining volatile compounds has given rise to variants: (i) solvent-free microwave extraction (SFME), which is based on applying the ability to generate microwave heating to perform a dry distillation, taking advantage of the intrinsic humidity of the biological material. However, the process usually requires more heating than hydrodistillation or steam stripping; it has been observed that it favors a more significant accumulation of oxygenated terpenoids compared to hydrodistillation (Lucchesi et al. 2004; Uysal et al. 2010); therefore, it could present organoleptic differences with respect to essential oils obtained by other methods: (ii) extraction by microwave steam distillation (MASD), where microwaves are used to irradiate the sample and generate the water vapor that will strip the compounds, either by hydrodistillation or direct stripping by steam (Drinić et al. 2020; Shang et al. 2020). (iii) Extraction by hydro-diffusion and microwave gravity, which is based on the SFME method, takes advantage of the humidity of the biological material to break the cells of the biological material and recover the compounds present in it. It is also coupled to a condensation system in the lower part of the system, which allows the extract to accumulate by gravity (Vian et al. 2008; Khan et al. 2016; Asofie et al. 2017).

The microwave-assisted extraction method is helpful for the recovery of volatile and non-volatile terpenoids, which is why it is one of the most applied non-conventional techniques in terpenoid recovery. Solvent-free microwave extraction is less common but has been shown to have a performance that rivals hydrodistillation and UAE (Lucchesi et al. 2004; Uysal et al. 2010; Liu et al. 2018). For example, Uysal et al. (2010) compared the extractability of laurel (*Laurus nobilis*) and lemon balm (*Melissa officinalis*) essential oils using SFME (50 min), hydrodistillation (150 min) and UAE (60 min). They observed that similar extract yields (mass/mass) were obtained with the three techniques for both organisms. Still, the proportion of oxygenated compounds

recovered with SFME is considerably higher than that obtained with hydrodistillation and UAE. Likewise, the study carried out by Liu et al. (2018) to obtain essential oil from *Cinnamomum camphora* observed that after optimizing the irradiation power and time parameters for the SFME method, an extraction time reduction of almost eight times was obtained compared to hydrodistillation, as well as a higher yield. When comparing the chemical profile of the composition of essential oils, the one obtained by hydrodistillation has 18 detected and identified compounds. In contrast, the one obtained by solvent-free microwave extraction recovers only 16 but has a more significant accumulation of oxygenated compounds, particularly camphor which is the main terpenoid of *C. camphora*.

In a recent study focused on essential oil recovery from an Italian variety of *C. sativa*, the extractability of the MAE method was compared to hydrodistillation. It was observed that using MAE there is a greater accumulation of (*E*)-caryophyllene, α -humulene, and cannabidiol. In comparison, hydrodistillation had a more significant accumulation of other terpenoids such as α -pinene, β -pinene, myrcene, limonene, 1,8-cineole, and terpinolene, among others. It should be noted that in this study, the conditions were optimized by principal component analysis to increase the accumulation of cannabidiol with MAE, and consequently, the recovery of other compounds was affected (Fiorini et al. 2020). In the study carried out by Drinić et al. (2020), the use of conventional hydrodistillation was compared with microwave-assisted hydrodistillation, evaluating three different levels of power supplied by microwaves (180, 360, and 600 W), and the extraction time was reduced, between almost 3 and 6 times, with respect to hydrodistillation depending on the power used. In addition, with the MAE method at 600 W, a higher yield of recovered essential oil and a chemical profile with more terpenoid derivatives were achieved. Another interesting observation was that oxygenated monoterpenoids accumulated preferentially when using the lower power levels with microwaves (180 and 360 W). At the same time, there was a more significant accumulation of monoterpenes when using hydrodistillation and MAE at 600 W. With the help of the parameter of different amount of supplied energy, the extraction can become selective toward the preferential accumulation of terpenes or oxygenated terpenoids.

In the extraction of non-volatile terpenoids, MAE has been evaluated in the recovery of triterpenic acids from olives, showing that the extraction using optimized microwave-assisted mashing achieves a higher recovery of oleanolic acid and maslinic acid, with a volume of solvent 15 times less than Soxhlet and only 10 min. In comparison, Soxhlet required 60 min for the process (Fernández-Pastor et al. 2017). Similarly, the study carried out by Kaur et al. (2019) showed that microwave-assisted extraction is more

efficient than the Soxhlet method and direct reflux, using aqueous ethanol (50%), for the recovery of ursolic acid, oleanolic acid, betulinic acid and lupeol from *Swerita* sp. On the other hand, when comparing extraction methods, with 50% ethanol, for the recovery of terpenoids from *Andrographis paniculata*, it was observed that the MAE method obtained the best result when recovering 14-deoxy-11,12-didehydroandrographolide but for the recovery of agrafolide and neoagrafolide, its recovery efficiency is lower than UAE and maceration at 80 °C (Chia et al. 2020).

Microwave-assisted extraction is more efficient than conventional methods, including Soxhlet, for terpenoid recovery. However, a factor that is decisive in enhancing the efficiency of the method is to carry out tests for the optimization of the power, temperature, and time parameters of the MAE, which can be achieved using statistical models such as principal component analysis (Fernández-Pastor et al. 2017; Chia et al. 2020; Drinić et al. 2020; Osorio-Tobón 2020).

Pulsating electric fields (PEF)

Extraction with pulsating electric fields is one of the unconventional methods that does not require generating heating for the recovery of compounds (Moreira et al. 2019). The method is based on generating pores in the cell membrane of the biological material using electrical pulses to facilitate the diffusion of the intracellular components toward the solvent. Among the main advantages of this method are the reduction of organic solvent residues and extraction time. It is helpful for the recovery of thermolabile compounds (Azmir et al. 2013; Bozinou et al. 2019; Vorobiev and Leblovka 2008). This method is not widely used in the recovery of terpenoids, but there are some recent reports in which the use of this method has been evaluated. A study was carried out to evaluate the effect of PEF on the extraction of α -acids, β -acids, and terpenoids from hops.

Regarding the terpenoids, it was observed that extraction assisted by electrical pulses (2.5 kV/cm, in methanol) recovered β -cubebene, β -cadinene, and δ -cadinene, which were not observed in methanol maceration. While with maceration, humulene epoxide and geranylisobutyrate were recovered, which were not observed using PEF; comparing the rest of the volatile terpenoid profile and its concentration, no significant differences were found between the samples treated with PEF and those not treated (Ntourtoglou et al. 2020). In another study, to obtain terpenoids from saffron using PEF (2 kV/cm), it was observed that using this method, there is a considerable loss of picrocrocin, safranal, and crocin terpenoids, affecting the quality of flavor, color, and aroma of saffron (*Crocus sativus*) (Neri et al. 2021). Another study focused on the recovery of terpenoids from rosemary (*Rosmarinus officinalis*) using electric field pulsations and maceration with ethanol at different concentrations

as solvent. After optimizing the ethanol concentration conditions and the electrical potential applied (25% ethanol, 20 kV, for 3 min), it was possible to recover approximately five times more oleanolic acid than in the maceration with agitation (50% ethanol for 9 min), as well as thymol almost ten times more. However, in this same study, other volatile terpenoids such as p-cymene, camphor, and carvacrol were lost with the use of PEF (Nutrizio et al. 2020).

Although the pulsating electric field extraction method is not a widely used strategy to obtain terpenoids and significant advantages in recovery have not been observed; for obtaining other compounds such as polyphenols, proteins, and polysaccharides, it is an extremely useful tool (Delsart et al. 2014; Ricci et al. 2020; Tzima et al. 2021). Therefore, it is necessary to continue investigating its application in the field of terpenoid recovery, optimizing the energy parameters and pulsation frequency to improve extractions.

Supercritical fluid extraction (SFE)

Due to the properties of supercritical fluids of diffusing like a gas and having relatively high densities, giving it a solvation capacity like that of liquids, they have been used for the extraction of secondary metabolites present in different types of samples (Azmir et al. 2013; Mahato et al. 2019; Osorio-Tobón 2020). Supercritical fluid extraction has turned out to be a very efficient method and is currently widely used. For secondary metabolites, CO₂ is used because its critical temperature is relatively low and can even be used to extract thermolabile compounds. The process consists of placing the sample of biological material in a chamber where the solvent will be delivered at its critical pressure and temperature to extract the intracellular compounds (Azwanida 2015; Osorio-Tobón 2020). They can also be used concomitantly with polar solvents such as ethanol and methanol, and this method has shown to be more efficient than decoction and maceration, also obtaining results in relatively short times. For these reasons, this method has become a popular alternative in recent years for the recovery of terpenoids (Bendif et al. 2018; Mushtaq et al. 2014; Uquiche et al. 2019; Uwin-eza et al. 2020).

Bendif et al. (2018) compared the chemical profile of the essential oil of two variants of *Thymus munbyanus* obtained by supercritical CO₂ fluid with that obtained with hydrodistillation, and the highest extract yield was obtained using SFE. However, the extracts obtained by SFE have a significantly lower number of volatile terpenoids compared to hydrodistillation, mainly recovering terpenoids of higher molecular mass, such as diterpenoids and triterpenoids. This same phenomenon has been observed in various studies on the recovery of essential oils from plants, where SFE does not recover the most volatile terpenoids (monoterpenoids), but those with the highest molecular mass, sesquiterpenoids

onwards, and oxygenates are the ones that mainly they accumulate (Conde-Hernández et al. 2017a; 2017b; Jokić et al. 2018; Kavoura et al. 2019). However, the optimization of pressure and temperature conditions (particularly 40 °C and 220 bar) of the extraction allows the recovery of the most volatile terpenoids, such as the case of eugenol from *Syzygium aromaticum* (Frohlich et al. 2019).

SFE extractability can also be enhanced using co-solvents, as observed when comparing SFE (CO₂), using 3% ethanol as co-solvent, with maceration, and UAE (with 96% ethanol) for acid recovery, acetyl-11-keto- β -boswellic from *Boswellia serrata*. After optimizing the pressure, flow, and extraction time (26 MPa, 3 mL/min, 225 min), in the supercritical fluid equipment, an extract yield of more than 2 times that obtained with UAE and maceration was obtained (Niphadkar and Rathod 2018).

Extraction with methods with pressure variations

To obtain compounds from natural sources, a series of strategies have been developed based on pressure variation when using solvents for extraction. Therefore, we can classify them into two main groups, those that use a vacuum (pressure reduction), and those that require a substantial increase in pressure (Chemat et al. 2020; Zhang et al. 2018).

Vacuum-assisted extraction Using the vacuum induces a forced extraction by passing the solvent faster through the biological material, which allows a greater drag capacity and reduction in process time (Borgarello et al. 2015; Wang et al. 2015; Wu et al. 2019). This technique is commonly coupled to extraction systems by reflux, Soxhlet, or even Clevenger equipment, with heating, the vacuum connection fits on the condenser to avoid loss of solvent and extract. Based on the same principle and structure of the system, extraction equipment reaching pressures close to 0.01 mmHg has been developed, and the process is called molecular distillation (Borgarello et al. 2015; Xiong et al. 2013; Zhang et al. 2018).

Borgarello et al. (2015) used this method to obtain oregano (*Origanum vulgare*) essential oil. These authors optimize the solvent flow at 0.3 mL/min and 30 °C with a pressure of 0.0031 bar, with which a greater recovery of terpenoids of the essential oil was achieved than that of the oil obtained by steam stripping. In another similar study, where oregano essential oil was also recovered, better recovery of terpenoids and particularly thymol was achieved with vacuum-assisted extraction compared to hydrodistillation. For this, parameters were optimized through a surface analysis of principal components, identifying that the optimal conditions for essential oil extraction were 0.558 bar, 253.8 min, and a 14:1 solvent ratio. Still, to favor the accumulation of thymol, the pressure of 0.3 bar is the most appropriate (Wu

et al. 2019). A variation of the method using an adsorbent (Tenax®) to capture the volatile compounds present in the headspace and forcing the extraction of those found within the matrix of the biological material. This method showed promising results, being much more efficient than hydrodistillation in the recovery of coriander essential oil. (Jeleń et al. 2021).

Rapid solid–liquid dynamic extraction (RSLDE) This is a relatively new method based on the Naviglio principle, and a Naviglio extractor is required for its application. In this equipment, the biological material is placed in a chamber where it will be immersed in the extraction solvent. Subsequently, pressure is exerted on the solvent, and the pressure is immediately released; this process is carried out using a piston mechanism; the repetitive pressure change forces the separation of intracellular molecules that do not have much interaction with the solid matrix of the biological material (Gallo et al. 2017, 2018; Naviglio 2003; Naviglio et al. 2019; Rocchetti et al. 2019; Palmieri et al. 2020).

Gallo et al. (2017) observed that RSLDE improves the recovery of the terpenoids stevioside and rebaudioside in less time than maceration (both cold and hot). This method was also tested to obtain essential oils of thyme, hemp, and coriander; showing to be more efficient than Soxhlet, UAE, and maceration for the recovery of volatile terpenoids in all three cases, but it required much more time than UAE for the process (2–6 h) (Palmieri et al. 2020).

Accelerated solvent extraction It is an accelerated extraction method that uses the solvent to recover the metabolites of interest from the biological material, applying pressures usually higher than 6 MPa and high temperatures (Chemat et al. 2020; Lv et al. 2010; Zhang et al. 2018). The utility of this method for the recovery of thermolabile compounds is still under discussion since some authors maintain the position that the decomposition of the compounds occurs due to the temperature. In contrast, due to the short periods of the process (less than 5 min), other authors suggest that it can be used to recover thermolabile molecules (Richter and Schellenberg 2007; Zhang et al. 2018). This process is entirely automated and can recover metabolites in very low concentrations (in the order of parts per billion) (Alexandre et al. 2017). Compared to MAE and UAE, ultrahigh-pressure solvent extraction has a higher yield in a shorter time (~2 min) to obtain the compounds (Alexandre et al. 2017; Guoping et al. 2012). Likewise, in the recovery of volatile terpenoids, there is a discussion about its efficiency since studies report that it is less efficient than maceration with agitation, hydrodistillation, and microextraction in the solid phase (Mardarowicz et al. 2004; Myers et al. 2021; Richter and Schellenberg 2007). Therefore, this method is more useful for recovering non-volatile terpenoids than volatile

ones, possibly by decomposition phenomena due to high pressures and temperatures.

Non-conventional solvents

Unconventional extraction solvents fall into two groups, ionic liquids, and deep eutectic solvents. Ionic liquids are saline compounds made up of organic cations and inorganic anions, which are naturally liquid or have a melting point below the boiling point of water (Ventura et al. 2017). These solvents have been very useful for the application in the extraction of secondary metabolites because the generation of polluting residues is minimal, in addition to presenting various advantages, such as the selective extraction of specific groups of secondary metabolites and being combined with practically all conventional and non-conventional extraction methods (Ventura et al. 2017; Xiao et al. 2018). For example, for the recovery of Paclitaxel from *Taxus x media*, different magnetic ionic liquids were tested in addition to UAE. After the optimization of the parameters, it was possible to define that the use of $[C_4MIM]FeCl_3Br$ at 1.2% in methanol with a 1:10.5 solid–liquid ratio and 30 min with ultrasonication is the best condition for the recovery of Paclitaxel (Tan et al. 2017). In addition, the use of ionic liquids that dissolve cellulose $[C_2MIM] [(MeO)(H)PO_2]$ allowed obtaining a triterpenoid never before described from *Ganoderma lucidum*, which was called ganoderic acid Σ (Murata et al. 2016). In the recovery of volatile terpenoids, they have also shown to be useful and optimizing the parameters, they show efficiency similar to extraction by steam stripping but not as efficient as reflux with organic solvents (Murata et al. 2017).

On the other hand, deep eutectic solvents are mixtures of Lewis acids and bases in the eutectic proportion; that is, the mixture has a lower melting point than pure compounds. Therefore, they will be found in a liquid state under the conditions of extraction (Smith et al. 2014). These solvents are so diverse that terpenoids such as menthol can even generate eutectic mixtures to extract other secondary metabolites (Jin et al. 2020; Wang et al. 2020b). In the recovery of terpenoids, the use of these solvents has been helpful in the extraction of cynaropicrin from *Cynara cardunculus*. Using the combination of decanoic acid: tetrabutylammonium chloride (2:1) in 70% water (%m/m) at 25 °C, and a 1:30 solid–liquid ratio, for 60 min of extraction, it was achieved a cynaropicrin recovery efficiency like that obtained with Soxhlet using chloroform and 7 h of processing (De Faria et al. 2017). In another study focused on the recovery of volatile terpenoids, it was observed that the choline chloride/glucose (5:2) system is the most efficient for the recovery of menthol, menthone, pulegone, and eucalyptol from mint leaves (Jeong et al. 2018). Likewise, these solvents have shown to be more efficient in the recovery of triterpenoids than the

Soxhlet method; for example, in the extraction of ursolic acid and oleanolic acid from *Eculaptus globulus*, extraction with eutectic solvents [thymol:menthol (2:1)] at 60 °C and 90 °C (72 h with agitation) achieved a recovery of more than double that obtained by Soxhlet with dichloromethane, during the same time (Silva et al. 2020).

Separation methods

The separation process represents the final step in the recovery of terpenoids and requires an appropriate selection of the method based on the characteristics of the compounds and the amount of the same that is desired to be recovered; this to have the least amount of possible losses in the process.

For purification, separation techniques allow fractions with less complexity to be obtained from the extract; subsequently, the process is complemented with polishing methods to obtain the pure compounds. The need to resort to polishing depends on the purposes of the analysis since, in some cases, the characterization of particular compounds can be obtained from the fractions without purifying the compounds.

Liquid–liquid extraction

The principle of liquid–liquid extraction is based on the solubility of each substance between two immiscible solvents, which gives rise to the concept of the partition and distribution coefficient, which takes into account the concentrations of the solute in each of the phases (Berk 2018). This technique has been beneficial in the enrichment of terpenoids, particularly in the case of terpenoids with carboxyl or amino functional groups with acid–base properties. Protocols are proposed that combine liquid–liquid partitioning methods to have a selective extraction, modifying the pH values in the aqueous phase (Ben Salha et al. 2021; Jiang et al. 2016). Although for the large-scale production of terpenoids, it is not a very profitable method due to the volumes of solvent required, as well as the separation and extraction processing time; liquid–liquid extraction is widely used in research (Hu et al. 2012; Troung et al. 2021; Wu et al. 2015; Zhang et al. 2018). In the recovery of fungal terpenic acids, the liquid–liquid extraction with chloroform and varying the pH with the use of $NaHCO_3$ (5%) is a very selective method for the recovery, later the neutral molecule is reconstituted by acidifying with HCl until $pH < 3$. Finally, it is re-extracted with chloroform to obtain an enriched fraction (Zhang et al. 2018). To reduce the volumes of solvent and make the extraction process more efficient, microextraction methods have been developed for the recovery of triterpenic acids of vegetable origin, using ultrasonication and ultracentrifugation, applying the distribution principle (Wu et al. 2015).

Chromatographic methods

For separating compounds of natural origin, chromatographic methods have been crucial for developing increasingly efficient protocols. Adsorption chromatographic techniques are the most useful in the purification process of natural products and include column chromatography and liquid chromatography, particularly HPLC (high-performance liquid chromatography), UPLC (ultra-performance liquid chromatography), which can be coupled to different detectors and analyzers for the identification and quantification of the compounds in the mixture. In addition to these methods, countercurrent liquid–liquid separation (CCS) has turned out to be a very efficient method for separating secondary metabolites (Ben Salha et al. 2021; Friesen et al. 2015; Liu et al. 2015).

Adsorption chromatography

Adsorption chromatography uses a stationary phase, made up of matrices (resins) that interact with the compounds to be separated, and a mobile phase that drags the compounds through the stationary phase. Column chromatography continues to be one of the main tools for the primary

fractionation of extracts to obtain terpenes/terpenoids (Duan et al. 2020; Huang et al. 2016; Li et al. 2017; Xu et al. 2019). New highly selective chromatographic resins for terpenoids are currently available, which can be used for column chromatography, preparative plate, and liquid chromatography (Table 3). HPLC high-efficiency liquid chromatography is presently one of the most used separation methods for terpenoids and many other compounds of natural origin, by using semi-preparative chromatographies (Bogdanov et al. 2017; Guo et al. 2019; Pan et al. 2017; Park and Kwon 2018; Truong et al. 2021; Wu et al. 2015; Xu et al. 2019). The UPLC technique originated from the HPLC, improving the resistance of the columns, adsorbents, and pumps to withstand higher internal pressures of up to 1200 bar. This has revolutionized the obtaining of secondary metabolites from different natural sources because it has allowed the separation of the samples with higher resolution and shorter analysis time (2–10 min), as well as reducing the volumes of solvents required for elution (Gaudencio y Pereira 2014; Hubert et al. 2017). Currently, this technique has been used in conjunction with mass analyzers, which also, with advances, have allowed more precise and mainly quantitative analysis to be applied in the separation and determination of secondary metabolites in different types of samples, managing to

Table 3 Some resins used for the separation of terpenoids

Resin	Matrix	Polarity	Terpenes/terpenoids recovered	References
SM-2	Cross-linked polystyrene-divinylbenzene	Non-polar	β -caryophyllene (sesquiterpene)	Grozdev et al. (2019)
S-X12	Cross-linked polystyrene-divinylbenzene	Non-polar	β -caryophyllene	Grozdev et al. (2019)
Polygoprep C18	Silica gel-C ₁₈	Non-polar	β -caryophyllene Herpetopanone (diterpenoid) Cembranoids (diterpenoids)	Grozdev et al. (2019) Pan et al. (2003) Bogdanov et al. (2017)
Rensa® PY	Cross-linked polystyrene-divinylbenzene-pyridine	Polar	β -caryophyllene	Grozdev et al. (2019)
Rensa® RP	Cross-linked polystyrene-divinylbenzene	Non-polar	β -caryophyllene	Grozdev et al. (2019)
Silica gel 60	Silica gel	Polar	β -caryophyllene Monoterpenoids; Sesquiterpenes(oids)	Grozdev et al. (2019) Wajs-Bonikowska et al. (2017)
XAD-2	Cross-linked polystyrene-divinylbenzene	Non-polar	β -caryophyllene	Grozdev et al. (2019)
XAD-4	Polyaromatic	Non-polar	<i>cis</i> - γ -bisabolene	Aschenbrenner et al. (2016)
XAD-7	Polyacrylate	Medium polarity	β -caryophyllene Erythric acids A-E (meroterpenoids)	Grozdev et al. (2019) Hu et al. (2012)
D101	Cross-linked polystyrene	Non-polar	Nafterpines (meroterpenoids) Labdanes, megastigmanes Glycosylated diterpenes Ginkgolides	Park and Kwon (2018) Xu et al. (2019) Guo et al. (2019) Zhou et al. (2018)
Silica gel 60N	Silica gel (neutra)	Polar	sterpurene; 9-Alloaromadendrene (sesquiterpenes)	Nagamine et al. (2019)

quantify up to 3 ng/mL (Berthold et al. 2021; Koetz et al. 2020; Wu et al. 2020).

Countercurrent chromatography

The countercurrent liquid–liquid extraction method, or countercurrent chromatography, is based on the partitioning principle in liquid–liquid extraction between two immiscible solvents. The difference with respect to a conventional solvent pair extraction lies in the fact that in this method, the sample is dissolved in a first solvent ("A"); subsequently, a solvent ("B") is made to flow in the opposite direction to gravity, which is immiscible with "A". One of the advantages of this method compared to HPLC or UPLC is that much larger sample volumes can be processed, which, given the time ratio, is comparable to the use of HPLC, with a more considerable amount of sample processed. Like the conventional liquid–liquid extraction method, terpenoids with functional groups that present acid–base balances (carboxyl and amino) can be recovered, by varying the pH in the aqueous phase, and the terpenoid remains in the phase where it is more soluble. Additionally, this method allows the use of multiple solvents for the extraction process in one run, and these can be recirculated to make it a continuous and more efficient process (Spórna-Kucab et al. 2020; Yu et al. 2020). This technique has been used at the end of the purification process, performing polishing, and there are some optimized extraction systems for the recovery of terpenes/oids (Table 4) (Friesen et al. 2015; Liu et al. 2014, 2015; Marques et al. 2018; Spórna-Kucab et al. 2020; Yu et al. 2020).

Hyphenated techniques

The processes for obtaining natural products have evolved along with technological advances in the various extraction, separation, and analysis methodologies. For this, instruments are used that, using the different extraction and separation principles mentioned in this review, allow the recovery of the compounds of natural origin. Initially, the coupled techniques had the purpose of identifying the compounds present in a sample without the need to recover it later, for which it was common to find that the separation process (usually chromatographic such as gas or liquid chromatography), was coupled to equipment destructive analysis, such as flame ionization detectors or mass spectrometers. However, the current need to reduce the losses of molecules has generated the development of online strategies that use other types of coupled analyzers and detectors, such as diode arrays, nuclear magnetic resonance, and infrared spectroscopy. Currently, there are state-of-the-art equipment found in online systems, which has allowed the automation of the analysis processes, reducing the steps of human intervention in the process and the analysis times (Gaudêncio and Pereira 2014; Hubert et al. 2017).

The use of hyphenated techniques has also allowed the development of specific strategies for the search for compounds, called dereplicative analysis, whose function is to provide fast and reliable analysis strategies for selecting natural sources of active principles (Gaudêncio and Pereira 2014; Hubert et al. 2017). In addition, it is possible to monitor the quality control of the extracts and ensure their effectiveness for application in the pharmaceutical and food sectors.

Table 4 Some systems used for the separation of terpenoids by countercurrent chromatography

Terpenes/terpenoids type	Elution system	Terpenes/terpenoids recovered	References
Triterpenic acids (separation)	n-Hexane/ethyl acetate/hydroxypropyl- β -cyclodextrin 0.1 M (9:1:10)	Ursolic acid; oleanolic acid	Wang et al. (2020a)
Volatil terpenes and terpenoids	Petroleum ether/acetonitrile/acetone (4:3:1)	Camphor; (<i>E</i>)-nerolidol; bornyl acetate; camphene	Marques et al. (2018)
Diterpenoids and triterpenic acids	Petroleum ether/ethyl acetate/methanol/water (1:0.8:1.1:0.6)	3,7-Dioxo-tirucalla-8,24-dien-21-oic acid; 3 α -acetoxyl-7-oxo-tirucalla-8,24-dien-21-oic acid; boscartol A	Yu et al. (2017)
Saponins	Chloroform/ethyl acetate/methanol/water (3:1:3:2)	11-Keto- β -boswellic acid; acetyl-11-keto-boswellic acid	de Souza Figueiredo et al. (2017)
	Ethyl acetate/ ethanol/water (1:0.2:1)	Jububogenins; Ketodamaran saponins	
	Hexane/ethyl acetate/butanol/ethanol/water (1:6:1:1:6)	Olean type saponins	
Monoterpenoids	n-Heptane/methanol, ethyl acetate/water (5:2:5:2)	Linalool; terpinen-4-ol; α -terpineol	Skalicka-Woźniak et al. (2013)

Concluding remarks

Obtaining terpenes/terpenoids from natural sources continues to be widely used due to the complexity of artificially synthesizing these compounds. For this reason, it is necessary to select the most appropriate strategy to obtain these compounds from the fragmentation process and extraction and purification. For this, the chemical properties of the molecules of interest must be considered, particularly the sensitivity to thermal decomposition and volatility, to select the most suitable processes. Regarding drying, the convection oven shows to generate the slightest change in the profiles of extracted compounds, but still, it is necessary to carry out tests at different temperatures to ensure the quality of the biological material. Likewise, assisted drying methods, particularly microwaves and ultrasonication, are very efficient for the drying process because they reduce time and at low cost, generating minor damage to biological material.

On the other hand, the extraction process is a decisive step in obtaining extracts enriched in a particular group of compounds. Regarding terpenes, the method selection must first focus on whether to recover volatiles or non-volatiles. For the recovery of volatiles, the stripping methods by steam are the most useful for large-scale applications, while at the analytical level, the adsorption methods are the preferred ones. Regarding non-volatile recovery, the Soxhlet method continues to be the most widely used due to its high recovery efficiency. Still, nowadays, non-conventional solvents (eutectic and ionic) are sought to be implemented to a greater extent to reduce the use of potentially harmful organic solvents. Similarly, assisted extraction methods have significantly accelerated extraction processes, reducing extraction times from days to a few minutes, and can be combined with virtually all conventional extraction methods.

Chromatographic techniques remain the most efficient methods for separating and purifying naturally occurring compounds, particularly terpenoids. For this reason, developing different resins and more efficient elution systems continues to be an essential issue in obtaining compounds of natural origin. New systems continue to emerge that allow the selectivity of group compounds, reducing process times.

Finally, when the compound profile of the biological material is unknown, it is advisable to evaluate different drying methods based on different principles and be able to choose the most appropriate one for the study. And in these cases, coupled techniques are beneficial to characterize the compounds of potential interest and designing a particular protocol for their recovery.

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Declarations

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